ALEENA KHURSHID

AI/ML Engineer (Entry-Level) | Computer Vision | NLP | LLM Applications | Python
Mirpur, Pakistan | aleenakhurshid233@gmail.com | linkedin.com/in/aleena-khurshid-ai | github.com/Aleena-Khurshid

PROFESSIONAL SUMMARY

Computer Systems Engineering graduate (MUST, 2022–2026) building AI systems in Python. Hands-on projects in computer vision and NLP, including video analysis with YOLOv8 and OpenCV, text extraction with EasyOCR, and a RAG pipeline using Gemini and Groq APIs served through FastAPI. Seeking a remote, paid AI/ML internship to apply these skills to real projects and learn production practices.

TECHNICAL SKILLS

Programming: Python (intermediate), C++, MATLAB, Object-Oriented Programming
Machine Learning & Deep Learning: Machine Learning, Deep Learning, PyTorch, Logistic Regression, NLP Sentiment Analysis, Hugging Face Transformers (DistilBERT fine-tuning)
Computer Vision: OpenCV, YOLOv8 Object Detection, EasyOCR
LLMs & Generative AI: RAG Pipelines, Gemini API, Groq API, TF-IDF Retrieval, Cosine Similarity Search, Whisper (speech-to-text)
Backend & Tools: FastAPI, SQLAlchemy, Streamlit, Gradio, Docker, pytest, Git, GitHub, Google Colab, Jupyter Notebook

WORK EXPERIENCE

Machine Learning & Deep Learning Intern 2025 (3 months)
Corvit Systems, Rawalpindi

- Completed machine learning and deep learning tasks during a 3-month internship.
- Built a Roman Urdu sentiment classifier that labels customer messages as negative, neutral, or positive (see Projects).

PROJECTS

VideoGPT: Intelligent Video Surveillance System Final Year Project (Solo)
Python, YOLOv8, OpenCV, EasyOCR, Groq LLM, SQLAlchemy, Streamlit | github.com/Aleena-Khurshid/VideoGPT

- Built a CCTV analysis pipeline for vehicle detection (YOLOv8), license plate recognition (EasyOCR), and vehicle color detection.
- Built natural-language search that converts queries such as "show red cars in gate1" into database queries using a Groq LLM.
- Added Gmail alerts for flagged events, a SQLAlchemy database for camera and detection records, and a Streamlit interface.

Video RAG Assistant github.com/Aleena-Khurshid/video-rag-assistant
Python, FastAPI, Streamlit, Gemini API, TF-IDF, Docker, pytest

- Built a RAG assistant that answers questions about a video's transcript and cites the timestamp of each answer.
- Implemented SRT/VTT subtitle parsing, 30-second overlapping text chunks, TF-IDF retrieval with cosine similarity, and Gemini answer generation with an offline fallback.
- Evaluated retrieval on 30 self-written questions across 3 self-made transcripts: Hit@3 of 0.93 and MRR of 0.93.
- Added 45 pytest tests, a CI check that fails if retrieval quality drops, and a Dockerfile.

Roman Urdu Sentiment Classifier github.com/Aleena-Khurshid/roman-urdu-sentiment
Python, TF-IDF, Logistic Regression, DistilBERT, FastAPI, Gradio, pytest

- Trained on about 20,000 labeled Roman Urdu sentences (public UCI dataset) with text cleaning and stratified 80/10/10 train, validation, and test splits.
- Compared a TF-IDF + Logistic Regression baseline (macro F1 0.673) with a fine-tuned multilingual DistilBERT (macro F1 0.628); the simpler baseline performed better.
- Served the model through a FastAPI endpoint with input validation, added a Gradio demo, and wrote pytest tests.

EDUCATION

BS Computer Systems Engineering 2022 – 2026
Mirpur University of Science and Technology (MUST), Mirpur, Pakistan | CGPA: 3.0 / 4.0
Relevant Coursework: Image Processing and Analysis, Machine Learning, Object-Oriented Programming, Python, C++

LANGUAGES

Urdu: Fluent (speaking, reading, writing) | English: Professional working proficiency (speaking, reading, writing)